Power Supply Systems

Part II

Phase 2: Conductors and Comparison of Systems

Aim: To understand conductors, stranded conductors, ACSR, AC/DC comparison, overhead/underground basis, conductor-volume comparison and semester-style derivations.

Style: Simple English, SN ma'am style, clear diagram first, formula after physical meaning.

Target: Semester-question-ready theory, derivation and numerical foundation.

Subject:	Power Supply Systems
Section:	Part II
Phase:	2 – Conductors and System Comparison
Level:	Zero to semester-question-ready

Academic article style notes

Contents

1 Phase 2 Roadmap

Phase 2 Goal

In Phase 1, we learnt how power goes from substation to consumer. In Phase 2, we learn what type of conductor is used and how different systems are compared economically.

Conductors → Stranded Conductor → ACSR → AC/DC Comparison → Overhead/Underground Basis → Conduc

Memory Hook

Phase 2 answers two questions:

Which conductor should be used?

Which system uses less conductor material?

2 Why Conductors Are Important

Definition

A **conductor** is a material or wire used to carry electrical current from one point to another.

In a power distribution system, conductor cost is very important because the length of lines is very large. So a good conductor should have:

1. low resistance,
2. good conductivity,
3. high mechanical strength,
4. low cost,
5. low weight,
6. good flexibility,
7. resistance against corrosion.

Semester Answer Style

If asked "What are the desirable properties of a conductor?", write the above points and explain each in one line.

3 Types of Conductors Used in Power Systems

Important conductors are:

1. Hard-drawn copper conductor
2. Hard-drawn aluminium conductor
3. Cadmium copper conductor
4. Steel-cored copper conductor
5. ACSR conductor

6. AAAC conductor
7. Stranded conductor

Conductor	Simple Use and Meaning
Hard-drawn copper	Good conductivity and strength, but costly and heavy.
Aluminium	Light and cheaper than copper, but conductivity is lower than copper.
Cadmium copper	Copper with small cadmium content; tensile strength increases.
Steel-cored copper	Copper strands around steel core; steel gives strength.
ACSR	Aluminium Conductor Steel Reinforced; most common for overhead lines.
AAAC	All Aluminium Alloy Conductor; good strength and corrosion resistance.
Stranded conductor	Made by twisting many thin wires together to increase flexibility.

4 Stranded Conductor

Definition

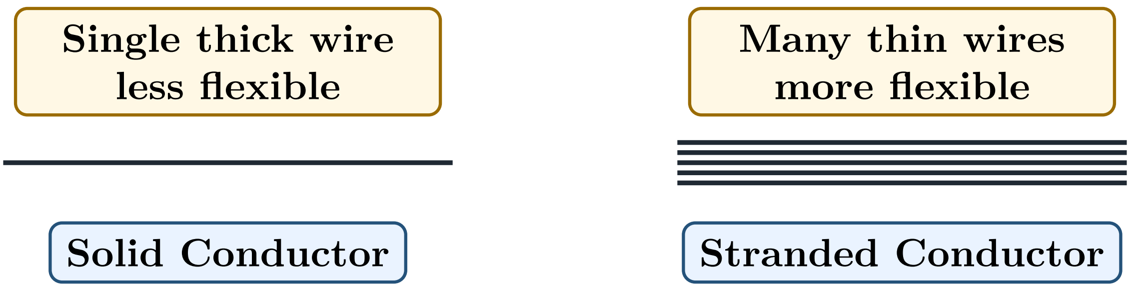
A **stranded conductor** is a conductor made by twisting several thin wires, called strands, together.

4.1 Why Stranding is Needed

If a long conductor is made as one solid wire, it becomes stiff and difficult to coil, transport and install. So many small wires are twisted together. This makes the conductor flexible.

Stranding increases flexibility.

4.2 Clean Diagram: Solid and Stranded Conductor



4.3 Important Facts

1. A stranded conductor has many small wires called strands.
2. The strands are arranged in layers.
3. Strands in successive layers are twisted in opposite directions.
4. Flexibility increases as number of strands increases.

5. For same total cross-section, stranded conductor has slightly higher resistance than solid conductor due to longer helical path.

4.4 Diameter Formula

For a stranded conductor,

$$\boxed{D = (2x - 1)d}$$

where,

$$D = \text{overall diameter of conductor}$$

$$x = \text{number of layers}$$

$$d = \text{diameter of each strand}$$

4.5 Meaning of Cable Notation

$$\boxed{19/0.64}$$

means:

- number of strands = 19,
- diameter of each strand = 0.64 mm.

Similarly,

$$\boxed{23/0.3}$$

means:

- number of strands = 23,
- diameter of each strand = 0.3 mm.

Semester Answer Style

Most probable question: What do you understand by stranded conductor? What are its advantages?
Answer: Give definition, diagram, need of flexibility, formula $D = (2x - 1)d$, and advantages.

5 Advantages of Stranded Conductors

1. They are more flexible.
2. They can be coiled and transported easily.
3. They are easier to handle during installation.
4. Mechanical strength can be increased by using a strong core.
5. Skin effect and inductance can be reduced by suitable construction.
6. They are suitable for overhead transmission and distribution lines.

6 ACSR Conductor

Definition

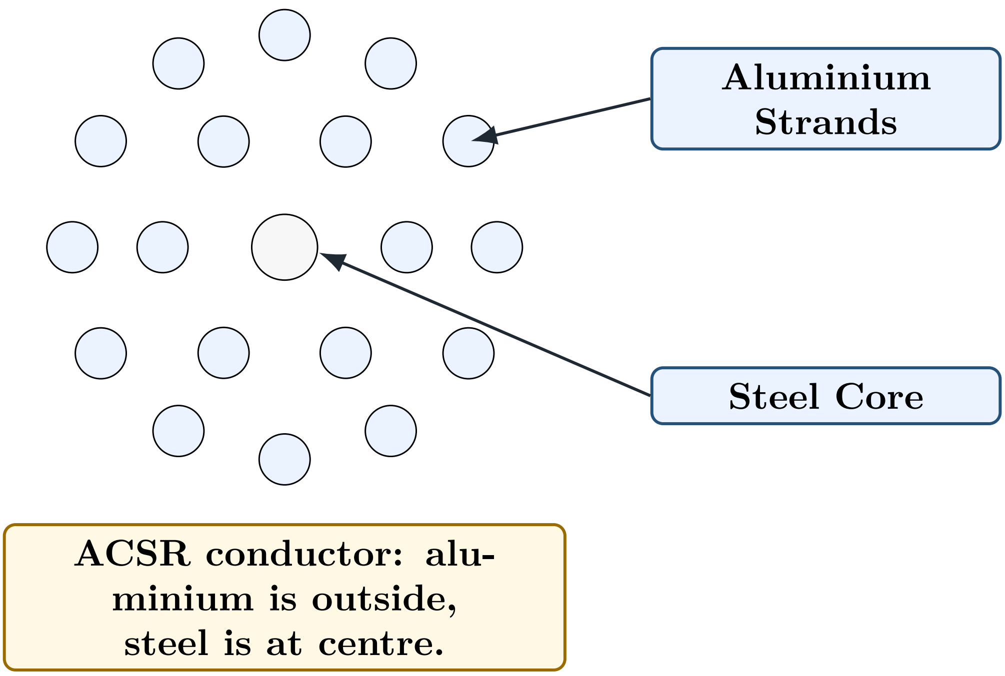
ACSR means **Aluminium Conductor Steel Reinforced**. It has a steel core at the centre and aluminium strands around the steel core.

6.1 Construction

- Steel is placed at the centre.
- Aluminium strands are placed around the steel core.
- Aluminium carries most of the current.
- Steel gives mechanical strength.

Aluminium carries current, steel gives strength.

6.2 Clean ACSR Diagram



6.3 Advantages of ACSR

1. It has high tensile strength due to steel core.
2. It is lighter than copper conductor.
3. It is cheaper than copper conductor.
4. It can be used for long spans.
5. Sag is reduced due to better mechanical strength.
6. Aluminium gives good conductivity.

6.4 Disadvantages of ACSR

1. Corrosion may occur if protection is poor.
2. Aluminium has lower conductivity than copper.
3. Joints need proper care.

Semester Answer Style

Semester answer style for ACSR:

1. Write full form.
2. Draw cross-sectional diagram.
3. Mention aluminium outside and steel core inside.
4. Explain function of each part.
5. Write advantages.

7 Comparison Between Copper and Aluminium

Point	Copper	Aluminium
Conductivity	Higher	Lower than copper
Weight	Heavy	Light
Cost	Costly	Cheaper
Mechanical strength	Good	Lower than copper
Use	Less used in overhead due to cost and weight	Widely used in overhead lines
Corrosion	Good resistance	Needs care in polluted areas

Memory Hook

Copper is a better conductor, but aluminium is lighter and cheaper. So aluminium and ACSR are very common in overhead lines.

8 AC Distribution and DC Distribution

8.1 DC Distribution

Definition

In **DC distribution**, electrical power is supplied using direct current.

8.1.1 Advantages

1. No reactance drop.
2. No skin effect.
3. Voltage drop calculation is simple.
4. No synchronising problem.

5. Less corona and interference for some cases.

8.1.2 Disadvantages

1. DC voltage cannot be directly stepped up or stepped down by transformer.
2. DC circuit breaking is difficult.
3. Conversion equipment is costly.
4. Generation at high DC voltage is difficult.

8.2 AC Distribution

Definition

In **AC distribution**, electrical power is supplied using alternating current.

8.2.1 Advantages

1. Voltage can be easily stepped up or stepped down by transformer.
2. AC generation is simple and economical.
3. AC is suitable for general power distribution.
4. Substation equipment is common and standard.

8.2.2 Disadvantages

1. Reactance drop is present.
2. Power factor affects current and losses.
3. Skin effect occurs.
4. Synchronising and stability problems may occur in large systems.

Semester Answer Style

Most probable question: Write the main advantages of AC transmission over DC transmission.
Key answer: AC voltage can be stepped up and down easily by transformer. This makes AC transmission and distribution economical and convenient.

9 Why High Voltage Reduces Conductor Material

For three-phase AC system:

$$P = \sqrt{3} V I \cos\phi$$

For same transmitted power P and same power factor,

$$I = \frac{P}{\sqrt{3} V \cos\phi}$$

So,

$$\boxed{I \propto \frac{1}{V}}$$

Copper loss is:

$$P_{\text{loss}} = 3I^2R$$

So,

$$\boxed{P_{\text{loss}} \propto I^2 \propto \frac{1}{V^2}}$$

Therefore, if transmission voltage increases:

$$\boxed{\text{Current decreases}}$$

$$\boxed{\text{Copper loss decreases}}$$

$$\boxed{\text{Conductor material required decreases}}$$

$$\boxed{\text{Transmission efficiency increases}}$$

Semester Answer Style

Semester statement: For the same power, when voltage is increased, current decreases. Therefore I^2R loss decreases and smaller conductor cross-section is sufficient. Hence conductor material decreases with increase in transmission voltage.

10 Systems of Power Transmission

Common systems are:

1. DC two-wire system
2. DC two-wire system with midpoint earthed
3. DC three-wire system
4. Single-phase AC two-wire system
5. Single-phase AC two-wire system with midpoint earthed
6. Single-phase AC three-wire system
7. Two-phase AC three-wire system
8. Two-phase AC four-wire system
9. Three-phase AC three-wire system
10. Three-phase AC four-wire system

Memory Hook

Semester questions mainly ask comparison of conductor material between two systems. So first learn the assumptions and method.

11 Basis of Comparison: Overhead and Underground Systems

This is a very important point.

11.1 For Overhead Lines

Definition

For **overhead transmission**, the basis of comparison is the maximum voltage between each conductor and earth.

Reason: Overhead conductors are supported on towers through insulators. The insulation requirement depends mainly on voltage between conductor and earth.

$$\boxed{\text{Overhead basis: voltage between conductor and earth}}$$

11.2 For Underground Cables

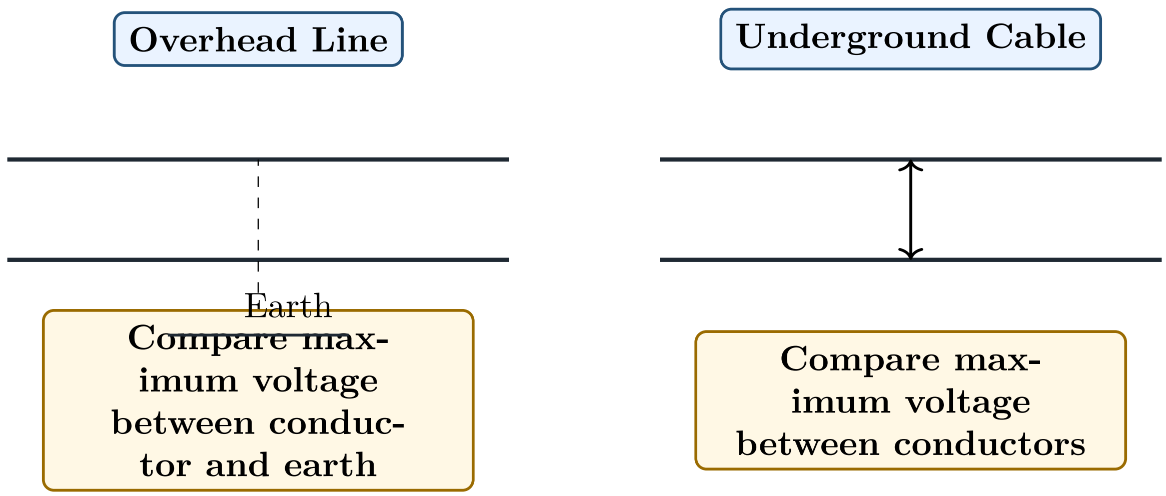
Definition

For **underground cables**, the basis of comparison is the maximum voltage between two conductors.

Reason: In underground cables, conductors are inside the same cable system. The dielectric stress depends mainly on voltage between conductors.

$$\boxed{\text{Underground basis: voltage between two conductors}}$$

11.3 Clean Basis Diagram



Common Mistake

Do not use the same voltage basis for overhead and underground questions. This is a common semester mistake.

12 General Assumptions for Conductor Material Comparison

Before deriving any comparison, write these assumptions:

1. Same power P is transmitted.
2. Same length l of line is used.
3. Same power loss P_{loss} is allowed.
4. Same transmission efficiency is maintained.
5. Same maximum voltage condition is used.
6. Same power factor $\cos\phi$ is used for AC systems.
7. Same conductor material is used.

Semester Answer Style

In every conductor-comparison answer, first write assumptions. This gives marks even before derivation.

13 Master Method for Conductor Volume Comparison

13.1 Step 1: Write Power Equation

For DC:

$$P = VI$$

For single-phase AC:

$$P = VI\cos\phi$$

For three-phase AC:

$$P = \sqrt{3}V_L I_L \cos\phi$$

13.2 Step 2: Write Loss Equation

$$P_{\text{loss}} = nI^2R$$

where n is the number of current-carrying conductors.

13.3 Step 3: Write Resistance

$$R = \frac{\rho l}{a}$$

where,

$$\rho = \text{resistivity}$$

$$l = \text{length}$$

$$a = \text{area of cross-section}$$

13.4 Step 4: Find Area

$$P_{\text{loss}} = nI^2 \frac{\rho l}{a}$$

$$\boxed{a = \frac{nI^2 \rho l}{P_{\text{loss}}}}$$

13.5 Step 5: Find Volume

$$\boxed{\text{Volume} = \text{number of conductors} \times a \times l}$$

Memory Hook

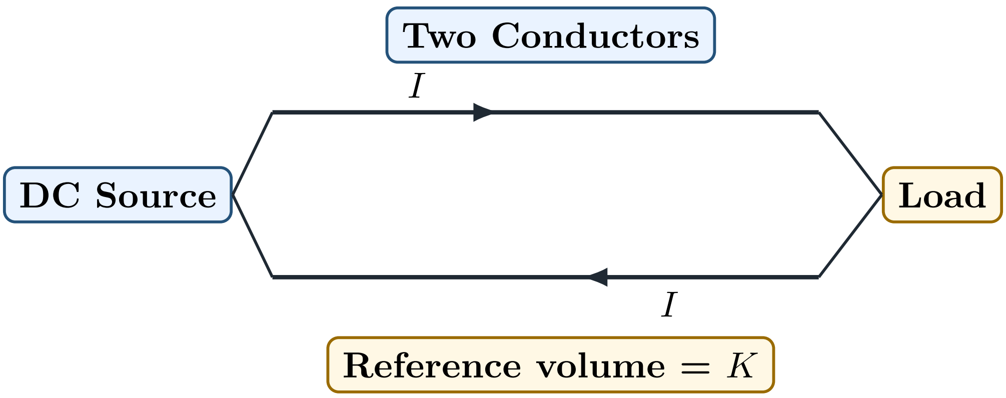
Conductor comparison is always:

$$\boxed{\text{Power} \rightarrow \text{Current} \rightarrow \text{Loss} \rightarrow \text{Area} \rightarrow \text{Volume}}$$

14 Reference Case: DC Two-Wire System

We take DC two-wire system as the basic case.

14.1 Diagram



14.2 Derivation

Let voltage between conductors be V_m.

$$P = V_m I$$

$$I = \frac{P}{V_m}$$

Loss in two conductors:

$$P_{\text{loss}} = 2I^2 R$$

$$P_{\text{loss}} = 2\left(\frac{P}{V_m}\right)^2 \frac{\rho l}{a}$$

So,

$$a = \frac{2P^2 \rho l}{P_{\text{loss}} V_m^2}$$

Volume:

$$V_c = 2al$$

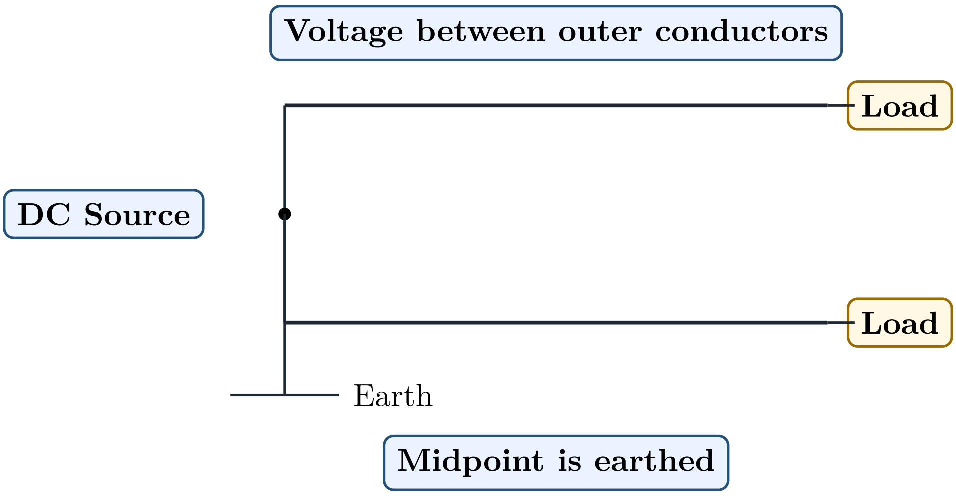
$$V_c = 2\left(\frac{2P^2 \rho l}{P_{\text{loss}} V_m^2}\right) l$$

$$\boxed{V_c = \frac{4P^2 \rho l^2}{P_{\text{loss}} V_m^2} = K}$$

This K is used as the basic comparison quantity.

15 DC Two-Wire System with Midpoint Earthed

15.1 Diagram



15.2 Main Result

For overhead system basis, this system requires less conductor material than the ordinary DC two-wire one-conductor-earthed case because maximum voltage to earth is reduced.

$$\boxed{\text{Volume} = \frac{K}{4}}$$

Semester Answer Style

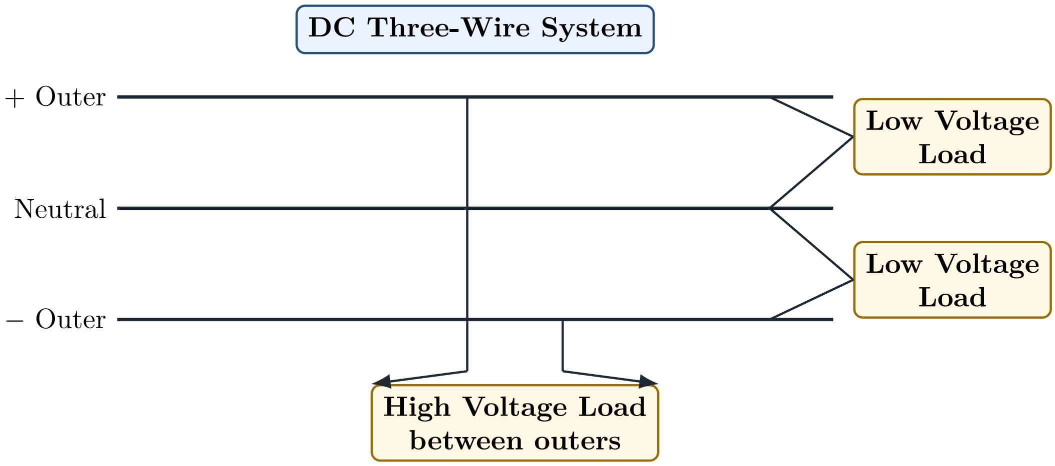
Write this only after stating the voltage basis clearly. For overhead comparison, basis is voltage to earth.

16 DC Three-Wire System

Definition

A **DC three-wire system** has two outer wires and one neutral wire. It can supply two voltages: V between outer and neutral, and $2V$ between the two outers.

16.1 Diagram



16.2 Important Points

1. It gives two voltage levels.
2. If loads on both sides are balanced, neutral current is zero.
3. If loads are unbalanced, neutral carries the difference current.
4. Neutral conductor is often taken as half the cross-section of each outer.

16.3 Volume Result for Underground Basis

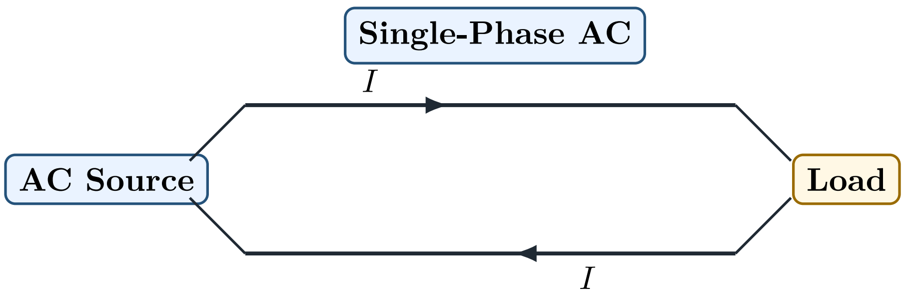
For underground comparison, if DC two-wire volume is K, then:

$$\boxed{\text{DC three-wire volume} = 1.25K = \frac{5K}{4}}$$

This result is useful in semester comparison questions involving underground cables.

17 Single-Phase AC Two-Wire System

17.1 Diagram



17.2 Main Formula

For single-phase AC:

$$P = VI\cos\phi$$

So,

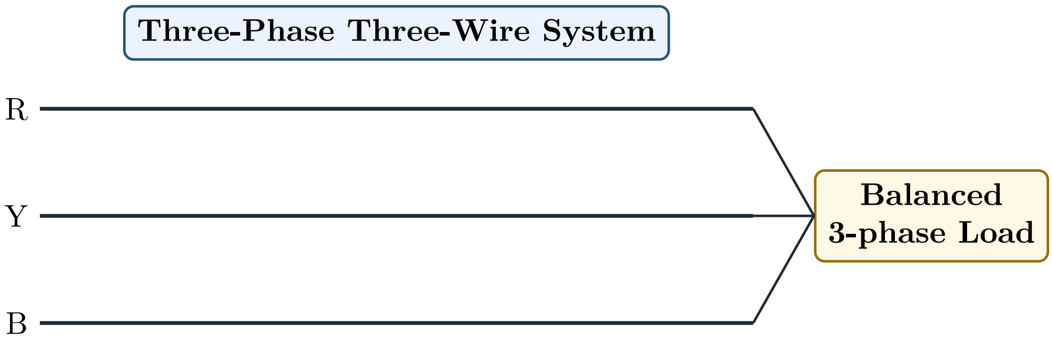
$$I = \frac{P}{V\cos\phi}$$

This means, if power factor is low, current increases.

$$\boxed{\text{Poor power factor} \Rightarrow \text{more current} \Rightarrow \text{more conductor material}}$$

18 Three-Phase Three-Wire System

18.1 Diagram



18.2 Main Formula

For three-phase AC:

$$P = \sqrt{3}V_L I_L\cos\phi$$

So,

$$I_L = \frac{P}{\sqrt{3}V_L\cos\phi}$$

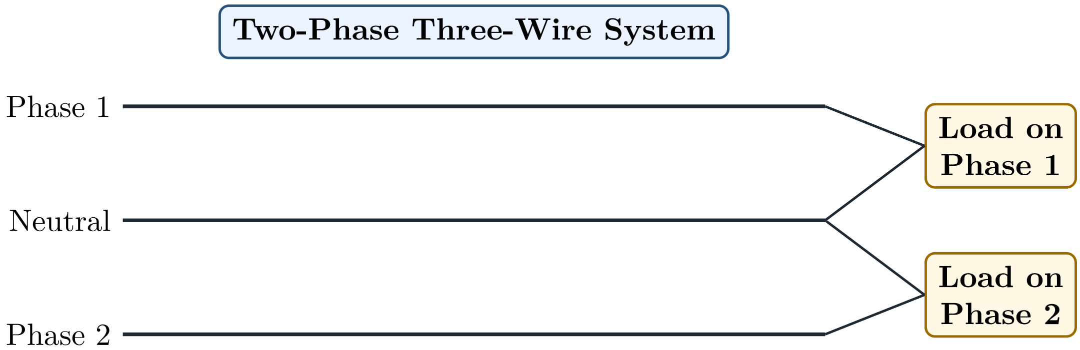
This system is economical for transmission because three-phase power transmission requires less conductor material than equivalent single-phase system for the same power and voltage condition.

19 Two-Phase Three-Wire System

Definition

A **two-phase three-wire system** has two phase conductors and one common neutral conductor. The two phase currents are normally displaced by 90°.

19.1 Clean Diagram



19.2 Important Point

The neutral current is the vector sum of the two phase currents. Since the two phase currents are 90° apart,

$$I_N = \sqrt{I^2 + I^2} = \sqrt{2}I$$

So the neutral conductor must be considered carefully.

20 Important Result Table for Conductor Volume

Let

$$K = \frac{4P^2\rho l^2}{P_{\text{loss}}V_m^2}$$

be the volume of conductor material required for the reference DC two-wire system.

20.1 Common Results

System	Overhead Basis	Underground Basis
DC two-wire, one conductor earthed	K	K
DC two-wire, midpoint earthed	$\frac{K}{4}$	K
DC three-wire system	$\frac{5K}{16}$	$\frac{5K}{4}$

Single-phase AC two-wire	$\dfrac{2K}{\cos^2\phi}$	$\dfrac{2K}{\cos^2\phi}$
Single-phase AC three-wire	$\dfrac{5K}{8\cos^2\phi}$	Usually derived as per voltage basis
Two-phase three-wire	$\dfrac{(3+2\sqrt{2})K}{8\cos^2\phi}$	$\dfrac{(3+2\sqrt{2})K}{4\cos^2\phi}$
Three-phase three-wire	$\dfrac{K}{2\cos^2\phi}$	Commonly treated by direct derivation

Common Mistake

Do not blindly memorise this table. In exam, first write assumptions and voltage basis. Then derive using power, loss and volume.

21 Semester Derivation: DC Three-Wire vs Two-Phase Three-Wire Underground Cable

This type is very important because it appears as a direct comparison question.

21.1 Given Condition

Compare:

DC three-wire system

with

AC two-phase three-wire system

for underground cables.

For underground cables, basis is:

$$\boxed{\text{same maximum voltage between two conductors}}$$

Let this voltage be V_m.

21.2 Part A: DC Three-Wire System

For balanced DC three-wire system:

$$P = V_m I$$

$$I = \frac{P}{V_m}$$

Loss occurs mainly in two outer conductors because neutral current is zero under balanced condition:

$$P_{\text{loss}} = 2I^2R$$

$$P_{\text{loss}} = 2\left(\frac{P}{V_m}\right)^2\frac{\rho l}{a}$$

So,

$$a = \frac{2P^2\rho l}{P_{\text{loss}}V_m^2}$$

If neutral area is half of outer conductor area,

$$\text{Total equivalent conductors} = 2 + \frac{1}{2} = 2.5$$

Volume:

$$V_{\text{DC 3-wire}} = 2.5al$$

$$V_{\text{DC 3-wire}} = 2.5\left(\frac{2P^2\rho l}{P_{\text{loss}}V_m^2}\right)l$$

$$\boxed{V_{\text{DC 3-wire}} = \frac{5P^2\rho l^2}{P_{\text{loss}}V_m^2}}$$

21.3 Part B: AC Two-Phase Three-Wire System

For two-phase three-wire system:

$$\text{phase voltage} = \frac{V_m}{\sqrt{2}}$$

Total power:

$$P = 2V_{\text{ph}}I\cos\phi$$

$$P = 2\left(\frac{V_m}{\sqrt{2}}\right)I\cos\phi$$

$$P = \sqrt{2}V_mI\cos\phi$$

So,

$$I = \frac{P}{\sqrt{2}V_m\cos\phi}$$

Neutral current:

$$I_N = \sqrt{2}I$$

If same current density is maintained, neutral area is:

$$a_N = \sqrt{2}a$$

Total equivalent conductor area:

$$2a + \sqrt{2}a = (2+\sqrt{2})a$$

Loss:

$$P_{\text{loss}} = (2+\sqrt{2})I^2\frac{\rho l}{a}$$

Substitute I:

$$P_{\text{loss}} = (2+\sqrt{2})\left(\frac{P}{\sqrt{2}V_m\cos\phi}\right)^2\frac{\rho l}{a}$$

$$P_{\text{loss}} = \frac{(2+\sqrt{2})P^2\rho l}{2V_m^2\cos^2\phi\; a}$$

Thus,

$$a = \frac{(2+\sqrt{2})P^2\rho l}{2P_{\text{loss}}V_m^2\cos^2\phi}$$

Volume:

$$V_{\text{2-ph 3-wire}} = (2+\sqrt{2})al$$

$$V_{\text{2-ph 3-wire}} = (2+\sqrt{2})\frac{(2+\sqrt{2})P^2\rho l^2}{2P_{\text{loss}}V_m^2\cos^2\phi}$$

$$\boxed{V_{\text{2-ph 3-wire}} = \frac{(3+2\sqrt{2})P^2\rho l^2}{P_{\text{loss}}V_m^2\cos^2\phi}}$$

21.4 Final Comparison

$$\frac{V_{\text{2-ph 3-wire}}}{V_{\text{DC 3-wire}}} = \frac{\dfrac{(3+2\sqrt{2})P^2\rho l^2}{P_{\text{loss}}V_m^2\cos^2\phi}}{\dfrac{5P^2\rho l^2}{P_{\text{loss}}V_m^2}}$$

$$\boxed{\frac{V_{\text{2-ph 3-wire}}}{V_{\text{DC 3-wire}}} = \frac{3+2\sqrt{2}}{5\cos^2\phi}}$$

Semester Answer Style

For this question, marks come from:

1. writing underground voltage basis,
2. drawing both systems,
3. finding current,
4. writing loss,
5. writing volume,
6. final ratio.

22 Semester Derivation: DC Two-Wire vs Two-Phase Three-Wire Overhead Line

This is also a repeated comparison family.

22.1 Given Condition

Compare:

DC two-wire system with one conductor earthed

with

AC two-phase three-wire system

For overhead lines:

$$\boxed{\text{maximum voltage between conductor and earth is same}}$$

Let this voltage be V_m.

22.2 Reference DC Two-Wire

As already derived,

$$\boxed{V_{\text{DC 2-wire}} = K}$$

where,

$$K = \frac{4P^2\rho l^2}{P_{\text{loss}} V_m^2}$$

22.3 AC Two-Phase Three-Wire

For overhead basis, phase-to-neutral voltage is V_m.

$$P = 2V_m I \cos\phi$$

$$I = \frac{P}{2V_m \cos\phi}$$

Neutral current:

$$I_N = \sqrt{2}I$$

Equivalent conductor area:

$$2a + \sqrt{2}a = (2 + \sqrt{2})a$$

Loss:

$$P_{\text{loss}} = (2 + \sqrt{2})I^2 \frac{\rho l}{a}$$

$$P_{\text{loss}} = (2+\sqrt{2})\left(\frac{P}{2V_m\cos\phi}\right)^2 \frac{\rho l}{a}$$

$$a = \frac{(2+\sqrt{2})P^2\rho l}{4P_{\text{loss}}V_m^2\cos^2\phi}$$

Volume:

$$V_{\text{2-ph 3-wire}} = (2+\sqrt{2})al$$

$$V_{\text{2-ph 3-wire}} = \frac{(2+\sqrt{2})^2P^2\rho l^2}{4P_{\text{loss}}V_m^2\cos^2\phi}$$

$$\boxed{V_{\text{2-ph 3-wire}} = \frac{(3+2\sqrt{2})K}{8\cos^2\phi}}$$

Semester Answer Style

This result is useful when the question says **overhead line**. Always mention that overhead comparison is based on voltage between conductor and earth.

23 Comparison of Distribution Systems

This part connects Phase 1 and Phase 2.

23.1 Radial System

- Simple and cheap.
- Fed from one end.
- Reliability is poor.
- Voltage drop is high at far end.

23.2 Ring-Main System

- Distributor forms a closed loop.
- More reliable than radial system.
- Supply can reach from two directions.
- Cost is higher than radial system.

23.3 Interconnected System

- More than one source feeds the network.
- Highest reliability.
- Protection is complicated.
- Cost is high.

Point	Radial	Ring Main	Interconnected
Cost	Low	Medium	High
Reliability	Low	Better	Highest
Voltage regulation	Poor	Better	Best
Protection	Simple	Medium	Complex
Use	Rural/small loads	Urban areas	Important dense load areas

24 Most Predictable Semester Questions from Phase 2

No.	Question	Marks
1	What do you understand by stranded conductor? What are its advantages?	5
2	Write short note on ACSR conductor with diagram.	5
3	What do you understand by 23/0.3 PVC cable?	2
4	Compare copper and aluminium conductors.	4
5	Write the advantages of AC transmission over DC transmission.	4
6	Show that conductor material decreases with increase in transmission voltage.	3–4
7	Explain the basis of conductor comparison for overhead and underground systems.	3–4
8	Compare conductor volume for DC three-wire and AC two-phase three-wire underground cable.	7–10
9	Compare conductor material for DC two-wire and two-phase three-wire overhead line.	7–10
10	Compare radial, ring-main and interconnected distribution systems.	6

25 Phase 2 Exam Answer Templates

25.1 Template 1: Stranded Conductor

Semester Answer Style

Answer sequence:

1. Definition.
2. Need of stranding.
3. Draw solid vs stranded conductor.
4. Write formula $D = (2x - 1)d$.
5. Explain 19/0.64 or 23/0.3 notation.
6. Advantages.

25.2 Template 2: ACSR Conductor

Semester Answer Style

Answer sequence:

1. Full form of ACSR.
2. Draw cross-sectional diagram.
3. Label steel core and aluminium strands.
4. Explain function of steel and aluminium.
5. Write advantages and use.

25.3 Template 3: Conductor Volume Comparison

Semester Answer Style

Answer sequence:

1. State assumptions.
2. State voltage basis: overhead or underground.
3. Draw the two systems.
4. Write power equation.
5. Write loss equation.
6. Find conductor area.
7. Find volume.
8. Take ratio and box final answer.

26 Phase 2 Final Memory Sheet

One-Page Memory Sheet

1. A conductor carries electrical current.
2. A good conductor has low resistance, good strength, low cost and good flexibility.
3. Stranded conductor is made by twisting many small wires.
4. Stranding increases flexibility.
5. $D = (2x - 1)d$ is the stranded conductor diameter formula.
6. ACSR means Aluminium Conductor Steel Reinforced.
7. In ACSR, aluminium carries current and steel gives strength.
8. Copper has better conductivity, but aluminium is cheaper and lighter.
9. AC is widely used because voltage transformation is easy.
10. High voltage reduces current, copper loss and conductor material.
11. Overhead comparison basis: voltage between conductor and earth.
12. Underground comparison basis: voltage between conductors.
13. Conductor comparison method:

$$\text{Power} \rightarrow \text{Current} \rightarrow \text{Loss} \rightarrow \text{Area} \rightarrow \text{Volume}$$

27 Phase 2 Self-Test

Answer these without seeing notes:

1. What is a stranded conductor?
2. Why are conductors stranded?
3. What is meant by 23/0.3 cable?
4. What is ACSR?
5. What is the function of steel in ACSR?
6. What is the function of aluminium in ACSR?
7. Why is AC distribution more common than DC distribution?
8. Why does high voltage reduce conductor material?
9. What is the comparison basis for overhead lines?
10. What is the comparison basis for underground cables?
11. Write the assumptions for conductor comparison.
12. Write the master method for conductor-volume comparison.

End of Phase 2

Next Phase:
Phase 3 – Kelvin's Law and Economic Size of Conductor
Annual cost, capital charge, cost of energy loss, Kelvin's law proof, cost curve and numericals.